Total No. of Questions : 8]

SEAT No. :

PC1971

[Total No. of Pages : 2

[6353]-306

T.E. (Computer Engineering)

HONORS IN DATA SCIENCE

Statistics and Machine Learning

(2019 Pattern) (Semester - II) (310503)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7or Q.8.*
2) *Neat diagram must be drawn wherever necessary.*
3) *Figures to the right indicate full marks.*
4) *Assume suitable data, if necessary.*

Q1) a) What is linear equation? What are the different methods to solve system of linear equation? Explain with suitable example. **[9]**

b) What is the difference between the Jacobian, Hessian and the gradient function? Explain with example the applications of each function. **[9]**

OR

Q2) a) Write a short note on: **[9]**

i) Partial derivative

ii) Multivariate calculus

b) What is the significance of chain rule in calculus? Explain chain rule with suitable Example. **[9]**

Q3) a) Draw and Explain Reinforcement Learning. Explain how does it work. **[9]**

b) Difference between supervised and unsupervised learning. **[8]**

OR

Q4) a) What are the different types of machine learning? **[9]**

b) What are the Applications, Perspective and Issues in Machine Learning. **[8]**

Q5) a) What is the role of cross-validation in evaluating a regression model? **[9]**

b) Explain the process of training a regression model using a dataset. **[9]**

OR

Q6) a) What do you mean by a linear regression? Which applications are best modeled by linear regression? **[6]**

b) Write a short note on: Simple Regression. **[6]**

c) Explain following Terms **[6]**

i) Cost function

ii) Gradient descent with respect to Multivariable regression

Q7) a) Define Bayes Theorem. Elaborate Naive Bayes Classifier working with example. **[9]**

b) Explain with example hypothesis space search in decision tree learning. **[8]**

OR

Q8) a) How to construct Decision trees? Write a short note on Classification of decision tress. **[8]**

b) Write short notes on: **[9]**

i) Bernoulli naive Bayes.

ii) Multinomial naive Bayes.

iii) Gaussian naive Bayes

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